



Research Paper

Immersed jet impingement heat transfer for servers in immersion cooling data center: A parametric study

Puhang Jin^a, Xue Zhang^b, Xiaoteng Zhang^a, Zhiming Tan^b, Gongnan Xie^{a,b,*}^a Ocean Institute, Northwestern Polytechnical University, Taicang 215400, China^b School of Marine Science and Technology, Northwestern Polytechnical University, Xi'an 710072, China

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ABSTRACT

Immersion and jet impingement cooling are effective solutions for addressing the thermal dissipation challenges of high-power electronic components in data centers. For the improvement of the thermal performance of hot-spots in immersion cooling servers, a jet-assisted immersion cooling is presented and studied. A numerical investigation considering the geometry of the heat sink, jet impingement height, mainstream flow rate, jet flow rate, and the distribution of jet orifices was carried out to unveil the convective heat transfer and flow field in immersed jet impingement cooling. The findings suggest that intermittent fins outperform continuous fins regarding thermal performance and flow resistance due to the unobstructed outflow, with the optimal jet heights being 2 mm and 2.5 mm. Throughout studied parameter ranges, larger mainstream and jet flow rates enhance thermal performance, albeit with a significant increase in pumping power consumption. The configuration of the jet orifices significantly improved temperature uniformity and saved pumping power consumption. Under the primary and jet flow rates of 1.31 L/min and 6.78 L/min, respectively. Specifically, under the mainstream and jet flow rates of 1.31 L/min and 6.78 L/min, respectively, the temperature difference on the heat source surface is reduced by 38.74%, and the pumping power is lowered by 7.81% when using Orifice Plate 2 versus the original uniformly distributed Orifice Plate 1. However, for elevated jet flow rates or longer liquid flow pipes, the internal pressure loss during the jet impingement has limited impact on the total pumping power. The present work provides a basic understanding of the immersed jet impingement cooling with cooling oils and serves as the guidance for addressing the local hotspot in immersion cooling servers with limited pumping power consumption.

1. Introduction

As the energy consumption of electronic devices increases and their size decreases, power density for a single server cabinet is elevated continuously, leading to the increment of thermal management demand in data center. Importantly, elevated temperatures not only degrade the performance of electronic devices but also significantly shorten their lifespan and diminish reliability [1]. Azarifar et al. [2] mentioned that with the continuous increase in heat flux, transitioning towards liquid cooling in data center cooling technology is an inevitable trend. Single-phase liquid cooling technology is regarded as a practical solution for addressing the heat dissipation challenges of increasing heat load in data centers [3–5].

Single-phase liquid cooling includes direct and indirect methods [6,7]. Immersion cooling falls within the scope of direct liquid cooling,

where electronic devices or components are completely submerged in a cooling medium to control temperature and dissipate heat effectively [8]. When electronic devices are fully immersed in a liquid, a tight thermal contact is formed between the liquid and the device surface, reducing contact thermal resistance. Immersion cooling has achieved some success in cooling personal computers (PC), decentralized computing servers and data centers [9]. Four types of working fluids can be used for this application, including deionized water, mineral oil, fluorinated liquids, and synthetic oils [10] where deionized water is barely used for its lower dielectric coefficient and difficulty in maintenance to keep the thermal properties unchanged. Fluorinated liquids and oils have stable chemical properties and excellent electrical insulation performance, effectively isolating electric currents to protect electronic devices from damage. While the fluorinated liquids and oils perform well at lower heat flux, their cooling efficiency decreases as heat flux rises, owing to higher viscosity and relatively poor thermal

* Corresponding author.

E-mail address: journals_conan@163.com (G. Xie).<https://doi.org/10.1016/j.applthermaleng.2025.126628>

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Nomenclature			
<i>Regular letters</i>		u_i, u_j	Velocities components along the x_i and x_j directions
C_p	Constant pressure specific heat capacity, J/(kg·K)	V	volume flow rate, m ³ /s
D	Hydraulic diameter, mm	V_1	Mainstream flow rate, m ³ /s
d	Diameter of pipe, mm	V_2	Jet flow rate, m ³ /s
d_n	Diameter of jet hole, mm	x_i, x_j	Cartesian coordinate system
e	Relative deviation, %	y	Distance to the nearest wall
f	Flow resistance coefficient	<i>Greek letters</i>	
F_1, F_2	Blending function	$\alpha_1, \alpha_2, \alpha_3$	Model parameter
H	Jet impingement height, mm	$\beta_1, \beta_2, \beta_3, \beta^*$	Model parameter
h	Heat transfer coefficient, W/(m ² ·K)	Δ	Amount of change
k	Turbulent kinetic energy, m ² /s ²	Δp	Pressure drop, Pa
l	Length of pipe, mm	Δp_p	Pressure drop in the pipe, Pa
L	Characteristic length, mm	ΔT	Temperature difference, K
p	Static pressure, Pa	λ	Thermal conductivity, W/(m·K)
P	Pumping power, W	μ, μ_t	Dynamic viscosity, Pa·s
P_d	Pumping power in the distributor, W	ν, ν_t	Viscosity, m ² /s
P_{jc}	Pumping power in the jet cavity, W	ρ	Fluid density, kg/m ³
P_{jet}	Pumping power in the computational domain, W	$\sigma, \sigma_{k1}, \sigma_{k2}, \sigma_{k3}, \sigma_{\omega 1}, \sigma_{\omega 2}, \sigma_{\omega 3}$	Surface tension coefficient, N/m
P_p	Pumping power in the pipe, W	ω	Unit dissipation rate, J/(kg·s)
Pr, Pr_t	Prandtl number	<i>Subscripts and superscript</i>	
p'_k	Production rate of turbulence	ave	Average
q	Heat flux, W/cm ²	f	Fluid
Re	Reynolds number	in	Inlet
S	Strain rate	i, j	Coordinates
T_{ave}	Average temperature of the heat source surface, K	max	Maximum
T_f	Average temperature of the fluid, K	min	Minimum
T_{in}	Fluid inlet temperature, K	out	Outlet
T_{max}	Maximum temperature of the heat source surface, K	<i>Abbreviation</i>	
T_{min}	Minimum temperature of the heat source surface, K	$CD_{k\omega}$	Position portion of cross-diffusion
T_{out}	Fluid outlet temperature, K		
u	Fluid velocity, m/s		

conductivity. To address this issue, common practices include increasing the cooling liquid flow and incorporating structured heat sinks.

Immersion cooling efficiency can be improved through the following approaches. Chen et al. [11] installed heat sinks on the CPUs to evaluate their performance of heat transfer. They discovered that the innovative three-dimensional horizontally arranged fin configuration exhibited better thermal performance compared to traditional designs. Cheng et al. [12] presented a groundbreaking cooling scheme that utilizes heat sinks and forced convection heat transfer. Simulation results indicate that a larger flow rate of the coolant leads to better heat transfer and lower CPU temperatures. However, with a lower flow rate, increased temperature and uneven temperature field were observed. Shah et al. [13] performed a numerical analysis to compare the performance of different heat sink geometries under single-phase oil immersion cooling conditions. They found that pin fin heat sinks and plate fin heat sinks showed no significant temperature difference relative to parallel plate heat sinks. Muneeswaran et al. [14] proposed a novel heat transfer enhancement scheme by installing fans in an immersion liquid cooling rack and analyzed the thermal performance variations caused by alternate fan configurations. The findings indicated that after arranging two fans in parallel, the thermal resistance decreased by 11%, and the chassis temperature dropped by 2.5 °C. For an immersion cooling data center, increasing the coolant flow rate can significantly raise power consumption of the system. While in an immersion cooling cabinet, apart from a few components like the CPUs and GPUs that generate substantial heat, the heat generated by most components is relatively low. Therefore, increasing the overall flow rate leads to unnecessary energy waste. The method of adding heat sinks can expand the surface area and enhance the cooling performance. Installing fans at the heat

sink to accelerate fluid flow and assist in the circulation of the coolant around high power components is a novel approach. However, the power consumption introduced by driving the fans has not been mentioned, making the cost-effectiveness of this method unclear.

Jet impingement cooling is a high-efficiency thermal management technology that utilizes high-speed fluid jets for heat dissipation. Based on the impingement configuration, jet impingement cooling is divided into three types: free jet, submerged jet, and submerged confined jet [15–17]. Free jet refers to the condition where fluid in the nozzle is different from the surrounding fluid, typically involving liquid from the nozzle being sprayed into a gas environment; submerged jet cooling refers to that the discharged coolant and the surrounding environment consist of the same fluid composition. While submerged confined jet cooling refers to the submerged jet with the external space constrained by solid walls. For active liquid cooling systems of electronic devices, where there is usually a requirement for compact cooling devices, often in confined spaces, jet impingement cooling in this field is mostly submerged confined jet. Based on fluid type and the temperature of the cooling surface, jet cooling is distinguished as single-phase and two-phase [18,19]. Due to the incomplete understanding of bubble dynamics and instability in heat transfer caused by bubbly flow in two-phase jet cooling, single phase jet impingement cooling is suitable for more real applications.

Womac et al. [20,21] have demonstrated that submerged impinging jets with $Re \geq 4000$ have superior heat transfer than free jet impingement, as a result of turbulence caused by the shear layer. While some scholars have demonstrated that the submerged confined jets show lower heat transfer coefficient compared to free jets, which is attributed to the recirculation of fluid heated by the target plate and entrainment

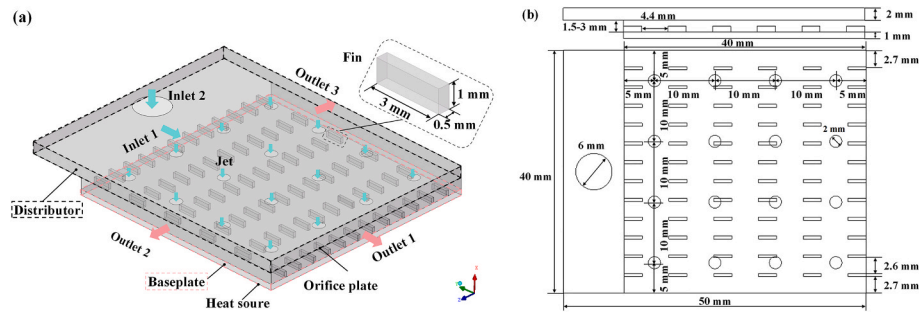


Fig. 1. Geometric parameters with intermittent fins (a) overall computational domain, and (b) front and top view of the heat sink and jet orifices.

into the impinging jet [22,23]. Huang et al. [24] suggest that in the $0 < r/d_n < 3$ region, the local Nusselt number- for the submerged confined jet cases are lower than those for the free jet case, while $3 < r/d_n < 5$, the submerged confined jet cases exhibit higher Nusselt number- than free jet case. According to Sarkar et al. [25], while submerged jets are suitable for electronic component immersion cooling, confined submerged jets prove more effective for jet-assisted heat sink configurations.

Typically, the nozzle axis is considered the centerline of the jet. The point where the jet centerline impinges on the cooling surface is termed the stagnation point. Maximum convective heat transfer Within the stagnation region, with the heat transfer coefficient decreasing sharply in the radial direction. Researches indicate that Nu is proportional to $Re^{(0.5-0.8)}$ when the jet diameter is fixed [26–29]. When the Reynolds number is constant, higher heat transfer coefficients are observed with a smaller jet diameter, especially in the stagnation region. This is because a reduction in jet diameter, under constant flow rate, results in increased jet velocity [26]. The jet-to-surface distance significantly influences the impingement heat transfer coefficient due to its impact on the potential core region. Robinson and Schnitzler [30] used water to study the effect of a submerged confined multiple jet with an H/d_n range of 2 to 30 on heat transfer performance, and Re range of 650 to 6,500. The results indicated that as H/d_n increases, the heat transfer coefficient typically exhibits a declining trend. The heat transfer coefficient exhibits minimal variation within the H/d_n is between 2 and 3, but undergoes sharp attenuation outside this optimal spacing interval. Tepe [31] also got similar conclusions, although the air-confined multiple jet was studied. In contrast, Sun et al. [32] and Tie et al. [33] concluded a different conclusion, they used water as the coolant. The former work on the submerged confined single jet ($0 \leq H/d_n \leq 20$) demonstrated optimal heat transfer at $H/d_n = 5$. The latter studied the effect of submerged confined multiple jets with an H/d_n range of 3.36 to 7 on thermal performance and found that the thermal performance is optimal at $H/d_n = 5$. Their different conclusions likely stem from variations in both the Reynolds number range and coolant type employed across studies.

For jet impingement, enhanced surfaces commonly induce better thermal performance compared to smooth surfaces. Attalla et al. [34] found that the heat transfer coefficient increases by 24% when using micro-cuboid fin surfaces instead of smooth surfaces. Wan et al. [35] created staggered and in-line configurations of square pin fin surfaces, achieving a 34.5% performance improvement compared to smooth surfaces. Chen et al. [36] conducted experimental tests in three scenarios: V-rib (ribs on both plates), V-rib-impingement (impingement plate only), and V-rib-target (target plate only). They found that the Nusselt number increased in both the V-rib-impingement and V-rib configurations, reaching a maximum enhancement ratio of 1.16. Alenezi et al. [37] found that surfaces with fins increased heat transfer coefficient by up to 15.6% compared to smooth surfaces. Cui et al. [38] demonstrated that while inclined fins enhance heat transfer in heat sinks, they simultaneously increase pressure drop. Their study further revealed that optimizing fin height and length provides more cost-effective performance improvements than adjusting tilt angle. However, structured surfaces may not guarantee enhanced thermal

performance in all cases. Yan et al. [39] examined the effects of staggered spacing and jet angles on thermal performance. Compared to the results without fins, the surface with the fins may enhance or deteriorate the heat transfer. Among the test angled-rib arrangements (45° , 60° , 90°), the 45° V-type ribs perform the best.

Although a single nozzle jet has strong heat dissipation capability, the sharp gradient in heat transfer coefficient can still cause a significant temperature difference on the cooled surface. As a result, jet array has become more widely used in the field of electronic device cooling [40]. Culun et al. [41] investigated the shape of jet holes, their configuration (in-line and staggered), the jet density in both transverse and longitudinal directions, and the different constraints affecting jet impingement heat transfer. Results indicate that square jet holes achieve optimal thermal transfer efficiency, while the jet arrangement is the least significant factor affecting heat transfer. Tian et al. [42] calculated the performance criteria η of air multi-jet with Re between 12,000 and 28,000. The calculation results show that staggered arrangements yield higher Nu than in-line arrangements. However, as the Re increases, the difference of Nu between staggered nozzle arrangement and inline nozzle arrangement becomes smaller. Zhang et al. [43] researched the effects of 6×6 , 5×6 , and 4×6 nozzle arrays on flow and heat transfer performance and the findings revealed that the 5×6 array exhibited the optimal comprehensive performance.

In data center cabinets, the heat generation of various components differs; for example, the CPUs and GPUs generate the most heat, while the others produce very little. Immersion liquid cooling can keep areas with lower heat generation within acceptable temperatures. While increasing flow rate in immersion liquid cooling can address high-power component thermal demands, excessive flow rates result in substantial pumping power penalties. Therefore, this paper proposes a single-phase jet-assisted immersion liquid cooling solution, where the entire cabinet employs immersion liquid cooling technology with small flow velocity, and jet impingement is applied specifically in the areas with high heat generation. The advantage of this method is that it improves the local convective heat transfer to meet high heat flux cooling demands without significantly increasing the overall pumping power consumption of the system. Specifically, in this paper, the main parameters of submerged jet impingement such as fin type, jet height, mainstream flow rate, jet flow rate, and jet array configuration were numerically investigated to reveal the underlying mechanisms that influence the thermal and flow characteristics in jet-assisted immersion cooling cabinet for high power density data center.

2. Numerical simulation methods

2.1. Geometric model

We Simulate the fluid flow and thermal performance of jet-assisted immersion cooling cabinet. We take the domain around a high power chip subjected to jet impingement to be the research object of this paper. Apart from the jets, the chip is also influenced by the main flow in the immersion cooling cabinet. So the chip's heat dissipation is indeed

Table 1
Physical properties of the cooling oil.

ρ (kg/m ³)	C_p (J/(kg·K))	λ (W/(m·K))	μ (kg/(m·s))
875	2052	0.189	0.0039375

realized by a submerged jet impingement cooling scheme.

The geometric model of calculation domain with intermittent fins is illustrated in Fig. 1(a), which includes a distributor and a heat sink. The distributor is 50 mm long, 40 mm wide, and 2 mm thick, and its bottom surface is an orifice plate. Jet orifices are configured in a 4×4 array, with each having a diameter of 2 mm. The center-to-center distance between jet orifices is 10 mm, while the center of jet orifices are positioned 5 mm from the perforated plate edge. To enhance both heat transfer area and flow turbulence, the baseplate (1 mm thickness) is designed with fins. The intermittent fins measure 3 mm length $\times$ 0.5 mm width $\times$ 1 mm height, arranged in a 12×6 array. Fin spacing is 2.6 mm, with 2.7 mm away from side fins to the baseplate edge along the Z-axis, while Y-axis spacing is 4.4 mm. The bottom surface with dimensions of 40×40 mm is acted as the heat source. Fig. 1(b) provides the specific dimensions of the components. The distance between distributor and base plate ranges from 1.5 mm to 3 mm in this study.

2.2. Governing equations

The model is simplified as follows: the cooling medium is considered as a continuous and incompressible fluid. Non-slip condition at the solid–fluid interface. Uniform velocity and temperature are assumed for the coolant at the inlets.

Continuity equation:

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (1)$$

Momentum equation:

$$u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[(\nu + \nu_t) \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right] \quad (2)$$

Energy equation:

$$u_i \frac{\partial T}{\partial x_i} = \frac{\partial}{\partial x_i} \left[\left(\frac{\nu}{Pr} + \frac{\nu_t}{Pr_t} \right) \frac{\partial T}{\partial x_i} \right] \quad (3)$$

where x_i and x_j denote Cartesian coordinates; u_i and u_j represent the corresponding velocity components in the x_i and x_j directions. The temperature, the pressure, and the fluid density are denoted by T , p , ρ , respectively. The turbulent kinematic viscosity is given by ν_t , Pr and Pr_t refer to the Prandtl number and turbulent Prandtl number, respectively.

The SST k - ω model simulates turbulent flow in this study. The equations governing the transfer of turbulent kinetic energy k and ω are given as:

$$\rho \frac{\partial (u_i k)}{\partial x_i} = \frac{\partial}{\partial x_i} \left[(\mu + \sigma_{k3} \mu_t) \frac{\partial k}{\partial x_i} \right] + p'_k - \beta' \rho k \omega \quad (4)$$

$$\rho \frac{\partial (u_i \omega)}{\partial x_i} = \frac{\partial}{\partial x_i} \left[(\mu + \sigma_{\omega 3} \mu_t) \frac{\partial \omega}{\partial x_i} \right] + 2(1 - F_1) \rho \sigma_{\omega 2} \frac{1}{\omega} \frac{\partial k}{\partial x_i} \frac{\partial \omega}{\partial x_i} + \alpha_3 \rho S^2 - \beta_3 \rho \omega^2 \quad (5)$$

where k represents the turbulent kinetic energy; p'_k denotes the turbulence production rate; ω is the specific dissipation; F_1 is the blending function; S represents the strain rate invariant.

The turbulence kinematic viscosity ν_t is denote as follows:

$$\nu_t = \frac{\alpha_1 k}{\max(\alpha_1 \omega, SF_2)} \quad (6)$$

The strain rate S is given by the following definition:

$$S = \sqrt{2 \left[\frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right] \left[\frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right]} \quad (7)$$

The F_1 and F_2 are defined as follows:

$$F_1 = \tanh \left(\left\{ \min \left[\max \left(\frac{\sqrt{k}}{\beta' \omega y}, \frac{500 \mu}{\rho y^2 \omega} \right), \frac{4 \rho \sigma_{\omega 2} k}{CD_{k\omega} y^2} \right] \right\}^4 \right) \quad (8)$$

$$F_2 = \tanh \left(\left[\max \left(\frac{2 \sqrt{k}}{\beta' \omega y}, \frac{500 \mu}{\rho y^2 \omega} \right) \right] \right) \quad (9)$$

$$CD_{k\omega} = \max \left(2 \rho \sigma_{\omega 2} \frac{1}{\omega} \frac{\partial k}{\partial x_i} \frac{\partial \omega}{\partial x_i}, 10^{-20} \right) \quad (10)$$

where y represents the closest wall distance.

The turbulence production rate p'_k is expressed as follows:

$$p'_k = \min \left(\mu_t \frac{\partial u_i}{\partial x_j} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right), 10 \cdot \beta' \rho k \omega \right) \quad (11)$$

The coefficients α_3 , β_3 , σ_{k3} , and $\sigma_{\omega 3}$ are specified:

$$\begin{cases} \alpha_3 = F_1 \alpha_1 + (1 - F_1) \alpha_2 \\ \beta_3 = F_1 \beta_1 + (1 - F_1) \beta_2 \\ \sigma_{k3} = F_1 \sigma_{k1} + (1 - F_1) \sigma_{k2} \\ \sigma_{\omega 3} = F_1 \sigma_{\omega 1} + (1 - F_1) \sigma_{\omega 2} \end{cases} \quad (12)$$

The constants for the turbulence models mentioned above are listed below: $\beta' = 0.09$, $\alpha_1 = 5/9$, $\beta_1 = 0.075$, $\alpha_2 = 0.44$, $\beta_2 = 0.0828$, $\sigma_{k1} = 0.85$, $\sigma_{k2} = 1$, $\sigma_{\omega 1} = 0.5$, $\sigma_{\omega 2} = 0.856$.

2.3. Boundary conditions and numerical methods

A constant heat flux condition is imposed on the heat source in the current study. The main flow (Inlet 1) and jet flow (Inlet 2) are configured with uniform velocity inlet boundary conditions. The inlet temperature is set as 298.15 K. The computational domain features pressure outlet boundaries (0 gauge pressure) on three sides for jet impingement, while the distributor walls are treated as adiabatic surfaces. The system consists of an aluminum solid and cooling oil coolant, whose physical properties are listed in Table 1.

Numerical simulations were conducted with commercial software ANSYS FLUENT. The pressure-based solver employs the SIMPLE algorithm for pressure–velocity coupling, with pressure discretization handled through the standard scheme for iterative solutions. The momentum equation and turbulent dissipation term are discretized using a second-order upwind scheme, while the energy equation is solved iteratively via the QUICK scheme. Convergence is determined when the residual of the continuity equation and the turbulent dissipation term in the momentum equation reach 10^{-3} , and the the energy equation residual falls below 10^{-6} .

2.4. Grid independence

Unstructured mesh was generated using ANSYS ICEM. The overall grid quality was maintained above 0.3 which is derived by the mesh quality function in ICEM. Boundary layer grids were refined near the walls and in the vicinity of the fins to ensure computational accuracy. Fig. 2 shows the overall and boundary layer mesh. To alleviate backflow at the outlets, the outlets were extended by 40 mm.

Four different meshes with varying grid densities were generated, with mesh number of 1.6 million, 4.0 million, 8.6 million, and 25.4 million, respectively. The simulation was conducted under the following operating conditions: inlet velocities of 0.5 m/s (Inlet 1) and 4 m/s (Inlet 2) with a uniform temperature of 298.15 K, combined with a heat flux boundary condition of 25 W/cm² applied to the heat source. The

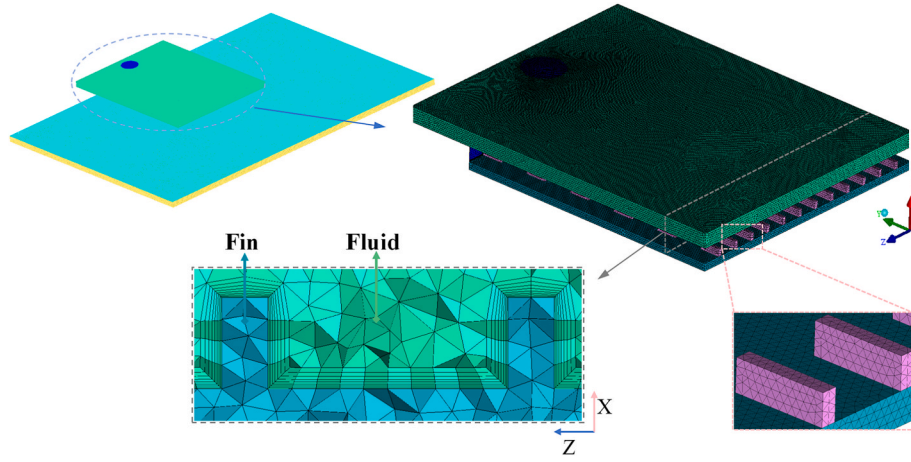


Fig. 2. Generated mesh for numerical simulations.

Table 2
Grid independence verification.

Mesh number	h (W/(m ² ·K))	e (%)	Δp (Pa)	e (%)
1.6 million	5322.73	4.07	4174.21	2.08
4.0 million	5254.49	2.74	4184.39	1.84
8.6 million	5171.63	1.12	4264.35	0.04
25.4 million	5114.51	0.00	4262.81	0.00

corresponding numerical results are summarized in Table 2. Taking the results of the case with 25.4 million as a reference, it can be observed that when the mesh number is 8.6 million, the relative deviations (e) for both the pressure drop and heat transfer coefficient remain below 1.2% which indicates that further mesh refinement beyond 8.6 million has a negligible impact on the computational results. Therefore, the mesh with 8.6 million elements was selected for subsequent calculations. The boundary layer mesh is applied near the fins and baseplate, with 8 grid layers for boundary layers. And the mesh configuration employs a first-layer height of 0.014 mm with a growth ratio of 1.2.

2.5. Model validation

For turbulent flow simulation using the RANS method, it is noted that while a turbulence model may reliably predict specific experimental results, its applicability may be limited to different flow conditions. In the literature, it has been found that in numerical studies on jet impingement flow, different turbulence models have been employed,

including the RNG $k-\epsilon$ model [44,45], Realizable $k-\epsilon$ model [46,47], and SST $k-\omega$ model [48]. To ensure the selection of an appropriate model, validation was performed against experimental data from similar configurations reported in Ref. [33]. Geometrical model of the jet array and heat sink in the referred experimental study is illustrated in Fig. 3(a). Deionized water was employed as the coolant at a constant inlet temperature of 294 K. The experimental conditions comprised a heat flux of 270 W/cm², jet diameters of 0.5–2.0 mm, and Reynolds numbers ranging from 3,500 to 12,000. It is noteworthy that the Reynolds number range investigated in this study spans from 5,300 to 8,000. For the simulation, we selected jet diameter of 1 mm, with distance of 7.0 mm between neighbouring jets, and 5×5 jet arrays configuration with 25 jets in total. H/d_n was maintained at 3.36 in model validation. Fig. 3 (b) compares numerical and experimental results regarding relative deviations of the heat transfer coefficient. The SST $k-\omega$ model demonstrates superior accuracy, showing the closest agreement with experimental results. Consequently, this model was selected for subsequent simulations.

2.6. Data reduction

The Reynolds number is determined by the expression:

$$Re = \frac{uL}{\mu/\rho} \quad (13)$$

where u denotes velocity, L is characteristic length, μ is dynamic viscosity, and ρ is fluid density.

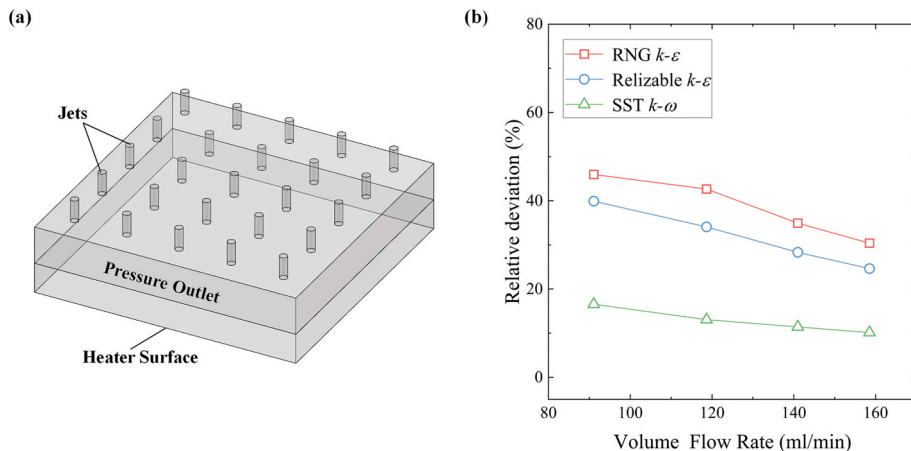


Fig. 3. (a) Geometrical model of the jet array and heat sink in the referred experimental study [33], and (b) Model validation against the experimental data [33].

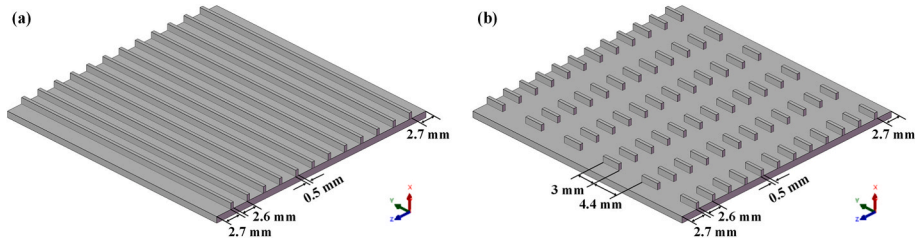


Fig. 4. Geometrical model of the heat sinks with (a) continuous, and (b) intermittent fins.

Table 3

Comparison between continuous fins and intermittent fins.

Fin type	T_{ave} (K)	T_{max} (K)	T_{min} (K)	ΔT (K)	Δp (Pa)	h (W/ (m ² ·K))
Continuous fins	375.43	388.57	366.38	22.19	10,657	2765
Intermittent fins	338.32	347.23	331.32	15.91	4264	5093

The heat transfer coefficient derives as:

$$h = \frac{q}{T_{max} - T_f} \quad (14)$$

where q denotes heat flux, T_{max} represents peak surface temperatures, and T_f indicates fluid temperature.

The fluid temperature derives as:

$$T_f = \frac{T_{in} + T_{out}}{2} \quad (15)$$

where T_{in} and T_{out} represent temperature of the inlet and outlet, respectively.

The heat source surface temperature difference is computed as:

$$\Delta T = T_{max} - T_{min} \quad (16)$$

where T_{min} is the heat source surface minimum temperature.

The pumping power consumption derives as:

$$P = V\Delta p \quad (17)$$

where Δp is pressure drop, V is volume flow rate.

The pressure drop from the inlet to the outlet is as follows:

$$\Delta p = p_{in} - p_{out} \quad (18)$$

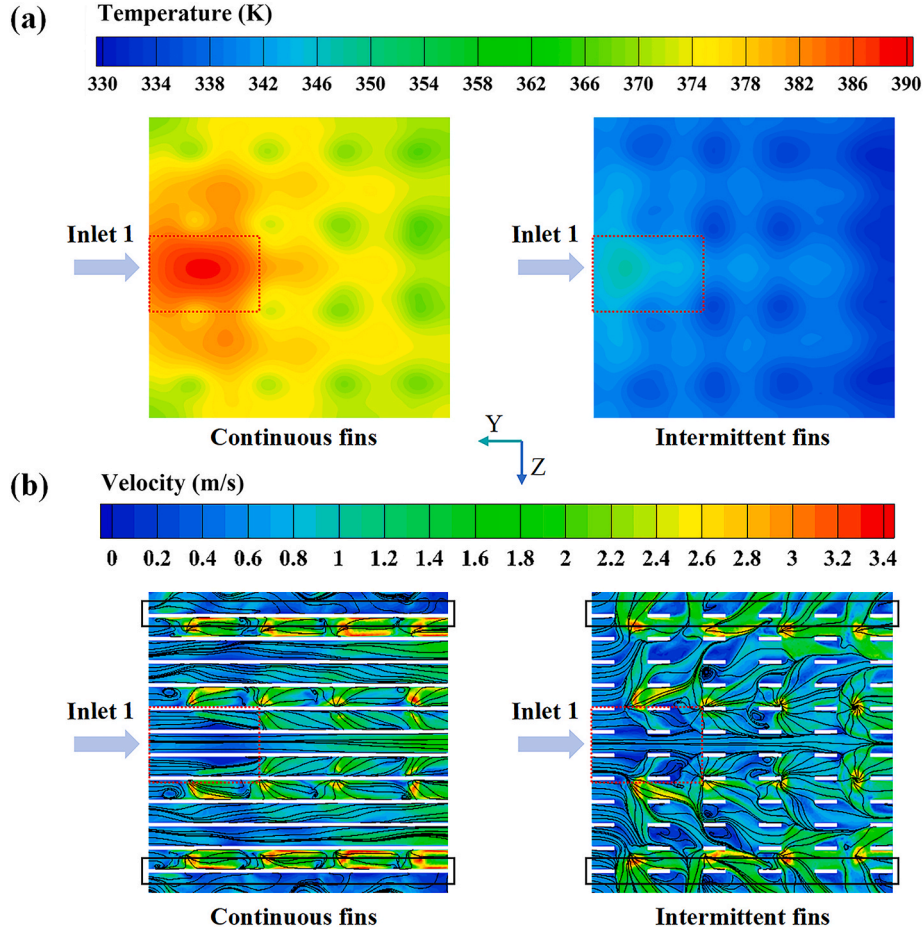


Fig. 5. (a) Temperature contour of the heat source surface, and (b) velocity contour and streamlines of the Y-Z cross-section 0.5 mm away from the baseplate inner surface (middle of the fin height).

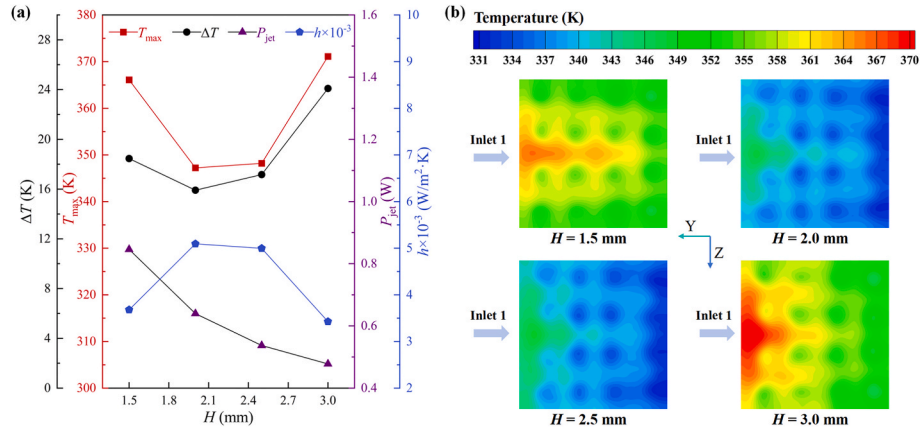


Fig. 6. (a) Variation trends of $T_{\max}$, ΔT , P_{jet} and h , and (b) temperature contours of heat source surface with four jet impingement heights H .

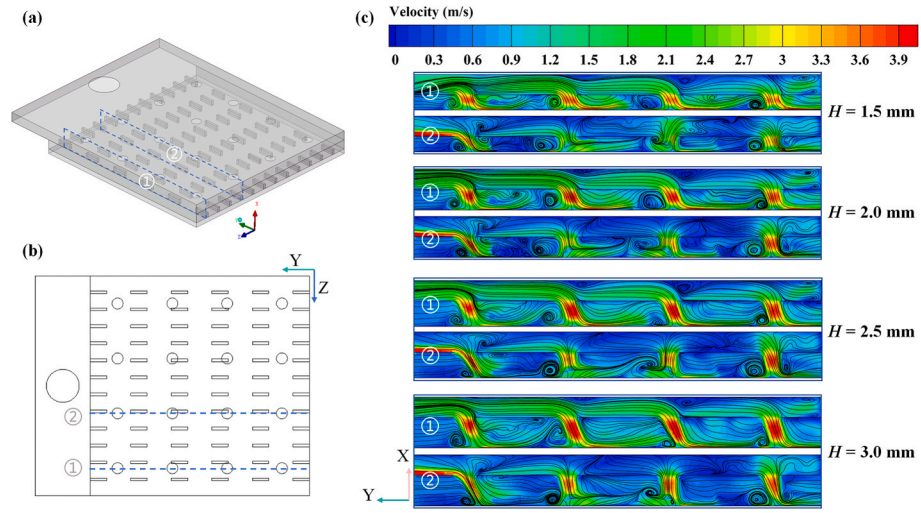


Fig. 7. (a) X-Y longitudinal cross section at position 1 and position 2 in 3D figure, (b) X-Y longitudinal cross section at position 1 and position 2 in 2D figure, and (c) velocity contours and streamlines of the X-Y cross section at two positions with four jet impingement heights.

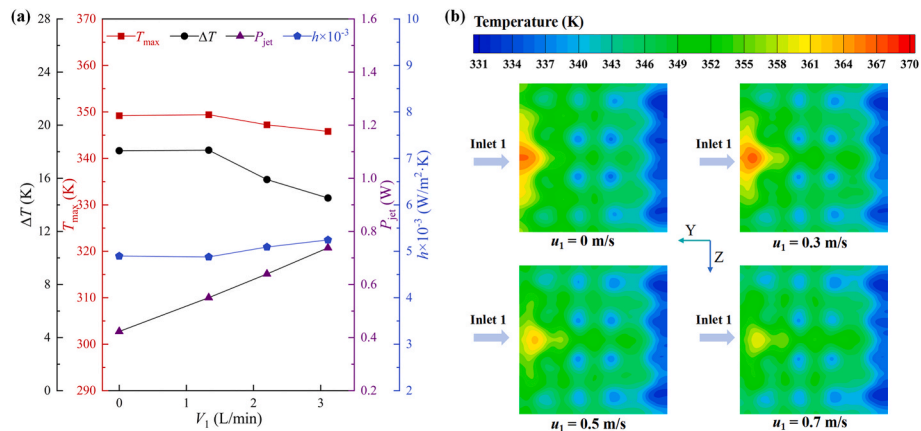


Fig. 8. (a) Variation trends of $T_{\max}$, ΔT , P_{jet} and h and (b) temperature contours of heat source surface with four different mainstream flow rates.

where p_{in} is the mass-weighted average pressure of inlet 1 and inlet 2, p_{out} is the mass-weighted average pressure of outlet 1, outlet 2 and outlet 3.

The mass-weighted average pressure of inlet 1 and inlet 2 is as follows:

$$p_{\text{in}} = \frac{m_{\text{in}1}p_{\text{in}1} + m_{\text{in}2}p_{\text{in}2}}{m_{\text{in}1} + m_{\text{in}2}} \quad (19)$$

where $m_{\text{in}1}$ and $m_{\text{in}2}$ are the mass flow rates at inlet 1 and inlet 2, respectively; $p_{\text{in}1}$ and $p_{\text{in}2}$ are the pressure at inlet 1 and inlet 2.

The mass-weighted average pressure of outlet 1, outlet 2 and outlet 3

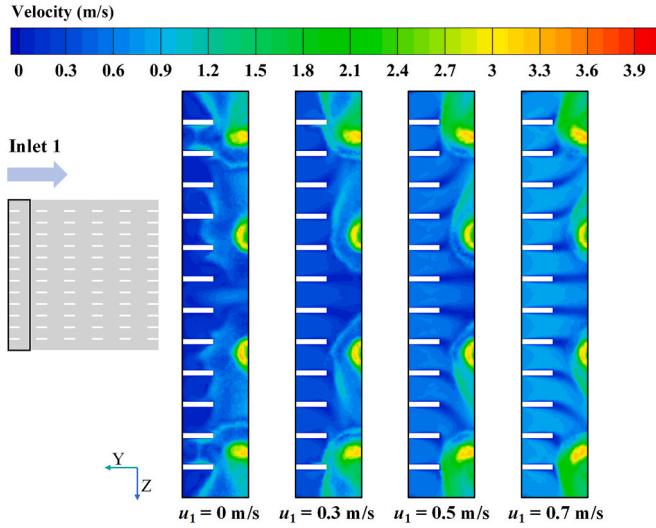


Fig. 9. Velocity contour of the Y-Z cross section 0.5 mm away from the baseplate inside the jet impingement domain at different mainstream flow rates.

Table 4
Numerical simulation results at two jet flow rates.

V_2 (L/min)	T_{ave} (K)	T_{max} (K)	ΔT (K)	h (W/(m ² ·K))	P_{jet} (W)
6.78	338.32	347.23	15.91	5093.32	0.64
10.18	330.89	334.96	10.14	6791.78	1.70

is as follows:

$$p_{out} = \frac{m_{out1}p_{out1} + m_{out2}p_{out2} + m_{out3}p_{out3}}{m_{out1} + m_{out2} + m_{out3}} \quad (20)$$

where m_{out1} , m_{out2} and m_{out3} are the mass flow rates at outlet 1, outlet 2 and outlet 3, respectively; p_{out1} , p_{out2} and p_{out3} are the pressure at outlet 1, outlet 2 and outlet 3.

The pressure drop in jet-assisted immersion cooling pipeline follows the empirical correlation of [49]:

$$\Delta p_p = f \frac{l \rho u^2}{2d} \quad (21)$$

where f is flow resistance coefficient, l is tube length.

The flow resistance coefficient can be represented by the following formula [49]:

$$f = \frac{0.3146}{Re^{0.25}} \quad (22)$$

3. Results and discussions

3.1. Effect of fin type

Two types of fins were studied in this section, i.e. continuous and intermittent fins shown in Fig. 4. Both fin types measure 0.5 mm in width and 1 mm in height. Heat sink with continuous fin includes 12 fins with a spacing of 2.6 mm, and the distance from the two side fins to the margin of the baseplate is 2.7 mm. The continuous fins were interrupted longitudinally to generate intermittent fins, which are positioned in a 12 × 6 array with distance of 4.4 mm. The simulations performed in this section have a jet impingement height of 2 mm, with jet orifices arranged in a 4 × 4 array. The velocity of Inlet 1 (mainflow) is 0.5 m/s (2.22 L/min) which is a normal liquid flow velocity in immersion cooling cabinet [12,14]. And the velocity of Inlet 2 is set to be 4 m/s (6.78 L/min), the resulting average jet velocity measured 2.25 m/s. A

298.15 K inlet temperature and 25 W/cm² heat flux were maintained on the heat source surface.

Table 3 presents the simulation results. Intermittent fins demonstrate markedly better performance than continuous fins, with a lower maximum heat source surface temperature T_{max} by 41.34 K, a lower average heat source surface temperature T_{ave} by 37.11 K, a smaller heat source surface temperature difference ΔT by 6.28 K, a lower pressure drop Δp by 6393 Pa (60% less), and a higher h by 2328 W/(m²·K) (84.2 % larger). The detailed comparisons of temperature and velocity fields between two types of fins are analyzed below.

Fig. 5 illustrates the temperature contours, along with the velocity contours and streamlines within the jet impingement domain. Compared to the continuous fin design, the intermittent fin exhibits a substantially lower heat source surface temperature. In both cases, the hot spots (red dashed rectangular frame) are located near the mainflow inlet (Inlet 1) where is hardly influenced by jet impingement. Within these areas, fluid velocities are significantly lower than in other locations, resulting in higher temperatures. The stagnation points of jet impingement exhibit lower temperature as expected and the area near the Outlet 1 shows the lowest temperature. This can be attributed to that in the downstream zone, particularly close to the outlets, the fluid velocity is higher than at other locations except for the stagnation region. The area enclosed by the solid black frame in the velocity contours and streamline indicate that the gaps between the intermittent fins facilitate fluid outflow from the side outlet, thus reducing the flow resistance.

3.2. Effect of jet impingement height

The literature review above reveals different points of view among researchers on how the height of jet impingement affects heat transfer. The data suggest that higher jet impingement heights result in diminished heat transfer efficiency [30,31], while others argue that there is an optimal jet impingement height that maximizes heat transfer performance [32,33]. Therefore, this section explores how jet height affects thermal and flow behaviors with respect to the proposed jet-assisted immersion cooling system. The jet impingement height studied includes 1.5, 2, 2.5, and 3 mm, while maintaining a mainstream flow velocity (Inlet 1) of 0.5 m/s, a jet flow rate (Inlet 2) of 6.78 L/min (the associated jet velocity is 2.25 m/s), and a constant heat flux of 25 W/cm² applied to the heat source surface.

Fig. 6(a) demonstrates that jet impingement height significantly affects heat dissipation performance and pumping power consumption. The variation trends between heat transfer performance (T_{max} , ΔT and h) and jet impingement height is not monotonous; instead, it exhibits an optimal range with small $T_{max}/\Delta T$ and large h , which is similar with the findings of Tie et al. [33] with plain surface. The case where the jet impingement height of 2.0 mm exhibits the lowest $T_{max}/\Delta T$ and highest h , followed closely by the case with a height of 2.5 mm. However, heat removal capability deteriorates sharply at jet impingement heights of both 1.5 mm and 3 mm. Nevertheless, the pumping power consumption increases monotonously by 77% with decreasing impingement height from 3 mm to 1.5 mm due to the narrowed flow pathways. Fig. 6(b) presents the heat source surface temperature contours. The left side represents the mainstream Inlet 1, while the other three sides represent the outlets. All four images share a common characteristic: high-temperature regions are concentrated near the central position of the mainstream Inlet 1, while low-temperature regions are located downstream near Outlet 1 which are similar to those in Fig. 5.

Fig. 7 illustrates heat sink velocity contours and streamlines at four jet impingement heights. The velocity contour in the X-Y crosssection were plotted along the central Y-axis between adjacent jet holes closest to Outlet 2. In X-Y longitudinal crosssection at position 1, the center of jet holes barely cover the fins, which means the jet impingement stagnation point mismatches with the fins. While in X-Y longitudinal crosssection at position 2, the impingement stagnation points of the two central holes partially intervene with the fins. With the same jet

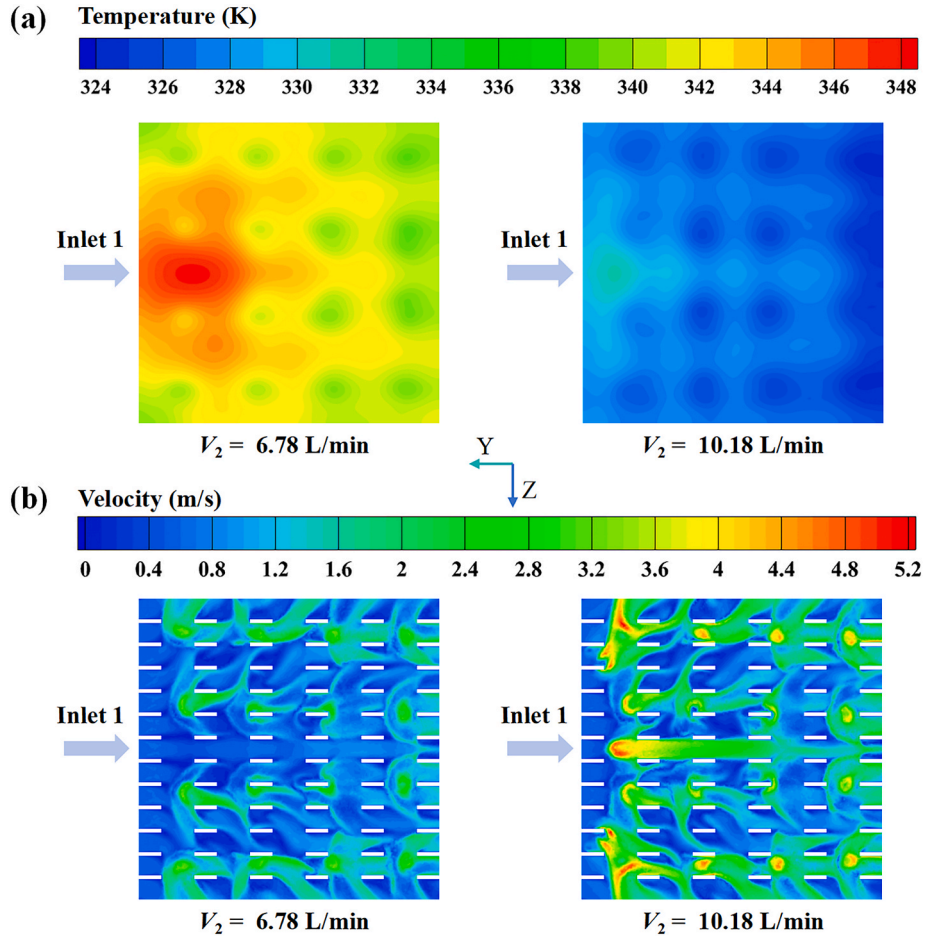


Fig. 10. (a) Temperature contour of the heat source surface and (b) velocity contour and streamlines of the Y-Z cross-section 0.5 mm away from the baseplate inner surface at two jet flow rates.

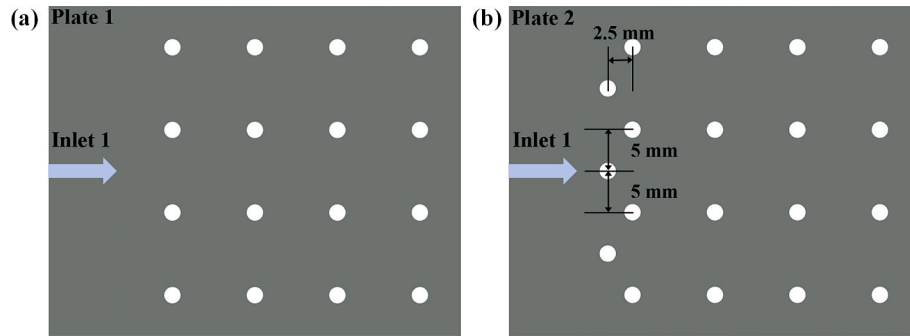


Fig. 11. Two orifice plate for jet distribution (a) original, (b) customized.

Table 5

Numerical simulation results with different orifice plates and mainstream flow rates.

Orifice plate	V_1 (L/min)	T_{ave} (K)	T_{max} (K)	ΔT (K)	h (W/(m ² ·K))	P_{jet} (W)
1	2.22	338.32	347.23	15.91	5093.32	0.55
2	2.22	339.65	343.76	10.11	5480.71	0.50
1	1.32	338.67	349.40	18.12	4877.89	0.64
2	1.32	340.24	344.88	11.10	5349.74	0.59

impingement height, the velocity at position 2 is significantly lower than at position 1, and the vortices at position 2 are also larger and more numerous than at position 1. This indicates that when the jet impacts the fins, the fins have a considerable obstructive effect on the fluid flow. As the jet impingement height decreases, this difference becomes more pronounced. Comparison of velocity profiles reveals that vortex formation in the jet chamber increases significantly with decreasing impingement height. When the jet impingement height is 1.5 mm, the vortices almost fill the entire jet chamber. These vortices may cause the fluid, which has already been heated by the bottom surface, to re-enter the jet stream, raising the jet temperature, in turn, affecting the cooling effectiveness. At a jet impingement height of 3 mm, the excessive flow distance leads to significant kinetic energy dissipation prior to surface

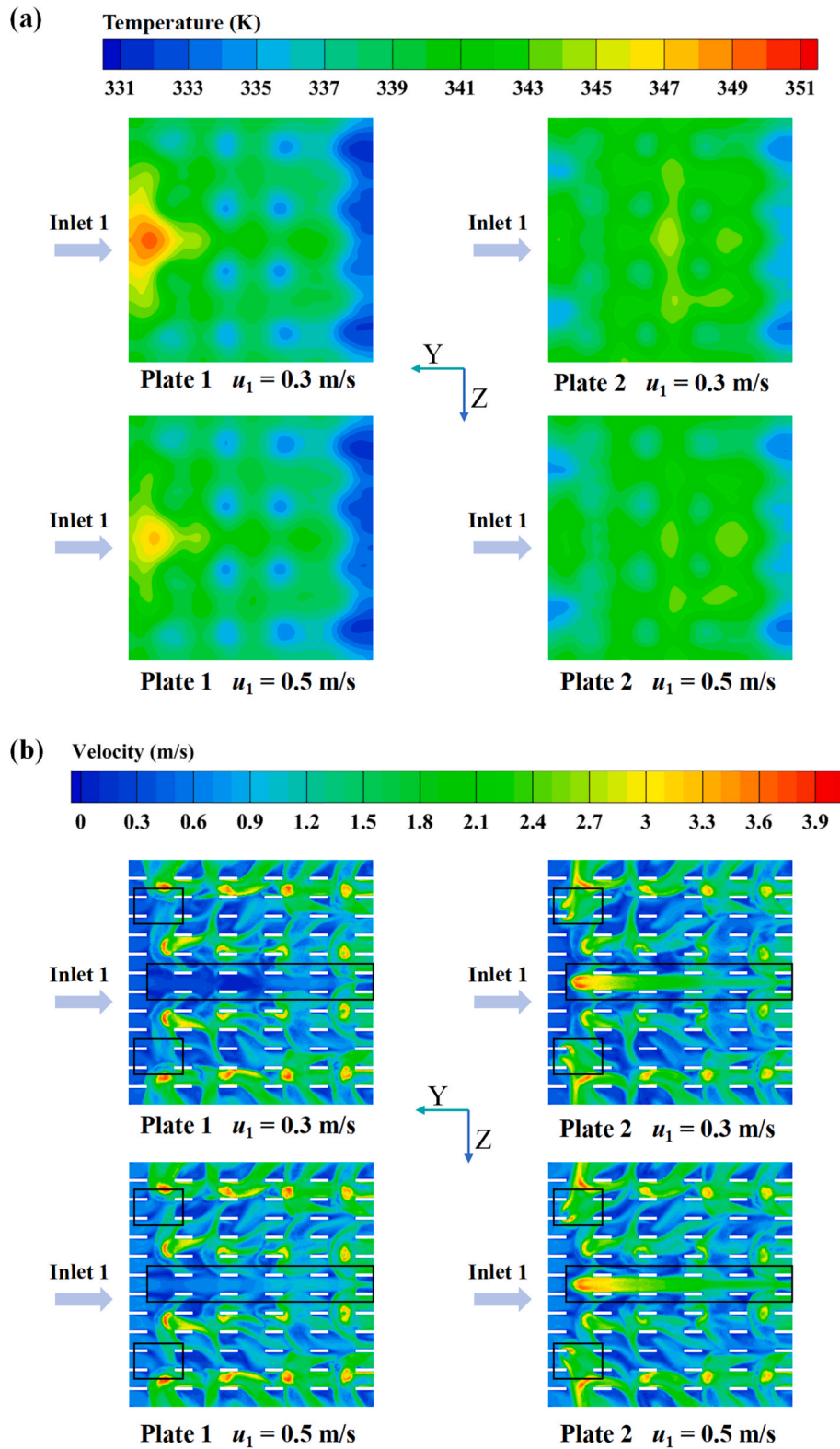


Fig. 12. (a) Temperature contours, and (b) velocity contours of the Y-Z cross section 0.5 mm away from the baseplate inner surface with two orifice plates and two mainstream flow rates.

impingement, resulting in degraded heat transfer performance. Furthermore, when there are multiple jets, the upstream jet affects the downstream jet, causing it to veer towards the outlet direction.

3.3. Effect of mainstream flow rate

As shown in Figs. 5 and 6, high-temperature areas are concentrated near the central position of the mainstream Inlet 1. We studied how mainstream flow rates affect thermal performance and flow behavior to

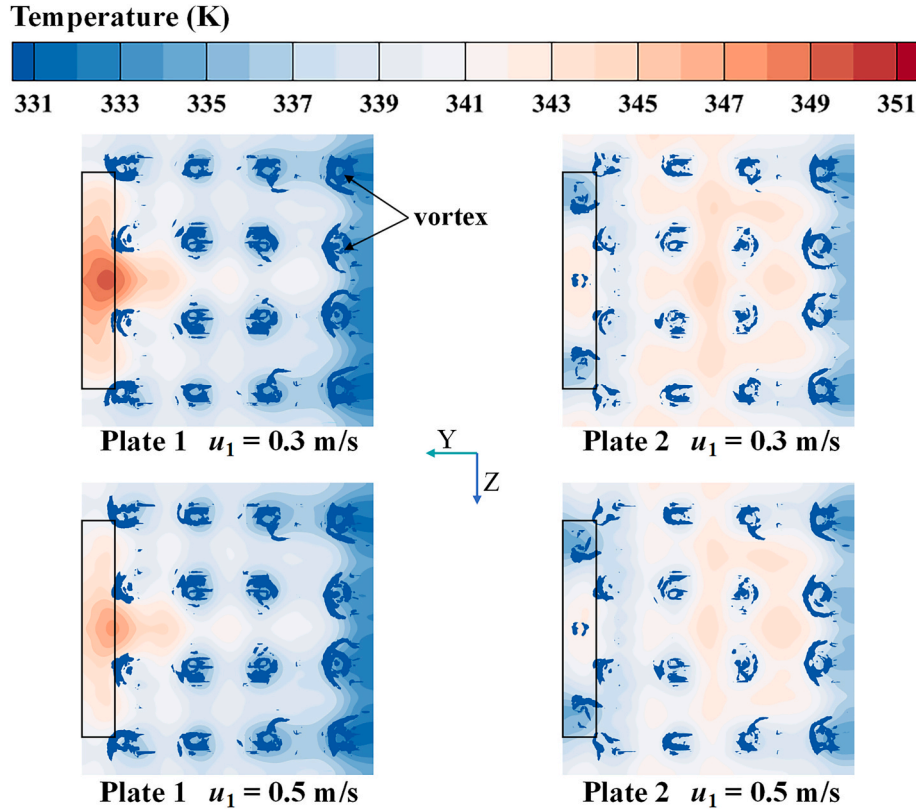


Fig. 13. Vorticity isosurface superposed with temperature.

Table 6
Operating conditions of 6 cases for pumping power consumption analysis.

Case	Orifice plate	H (mm)	Fin type	V_1 (L/min)	V_2 (L/min)
1	1	2	intermittent	2.2	6.78
2	1	2	continuous	2.2	6.78
3	1	3	intermittent	3.42	6.78
4	1	2	intermittent	3.11	6.78
5	1	2	intermittent	2.2	10.18
6	2	2	intermittent	2.2	6.78

enhance surface temperature uniformity. This section employs a 2 mm jet height with 25 W/cm² heat flux and 6.78 L/min flow rate (jet velocity = 2.25 m/s). We examined the performance of the heat sink at mainstream flow rates of 0 L/min, 1.32 L/min, 2.22 L/min, and 3.11 L/min, which correspond to mainstream velocities of 0 m/s, 0.3 m/s, 0.5 m/s, and 0.7 m/s, respectively.

Fig. 8(a) shows variation trends of $T_{\max}$, ΔT , P_{jet} and h . Results show that increasing the mainstream flow rate yields only marginal reductions in $T_{\max}$ and slight improvements in heat transfer coefficient, while more significantly reducing surface temperature difference. The increased mainstream velocity improves hotspot cooling near Inlet 1 (Figs. 5–6), explaining this behavior. However, the pumping power rises significantly. Specifically, when $V_1 = 3.11$ L/min, the heat source temperature reaches 345.82 K, only a reduction of 3.38 K compared to the case with no mainstream flow; the temperature difference decreases around 14.52 K; h reaches 5245.17 W/(m²·K), an increase of 347 W/m²·K (7.08% larger); and the pumping power increases to 0.74 W, a rise of 0.32 W (76.19% larger). Thus, increasing the mainstream is not an efficient method to enhance thermal performance due to the higher pumping power consumption. Fig. 8(b) demonstrates that increasing the flow rate reduces the high-temperature zone near Inlet 1 and improves bottom surface temperature uniformity. Notably, the hotspot persists near the mid-section of main flow Inlet 1. Therefore, the heat source surface

temperature uniformity can only be improved within a certain range by changing the main flow rate.

Fig. 9 shows the velocity contour on a horizontal cross-section 0.5 mm above the heat sink's baseplate inner surface at 4 different mainstream flow velocities. Clearly, with the mainstream flow rate rises, the velocity around the mainstream inlet (denoted by the solid black box) significantly increases, improving heat transfer while lowering local temperatures.

3.4. Effect of jet flow rate

This section examines the effects of jet flow rate on heat sink performance. The study employs a fixed jet height of 2 mm under 25 W/cm² heat flux and 2.2 L/min mainstream flow. Two jet flow conditions are analyzed: 6.78 L/min (jet velocity = 2.25 m/s) and 10.18 L/min (jet velocity = 3.37 m/s), comparing their fluid dynamics and thermal characteristics.

The numerical simulation results are summarized in Table 4. Increasing the jet flow rate reduces the heat source temperatures by: 12.27 K ($T_{\max}$), 7.43 K (T_{ave}), and 5.77 K (ΔT). This enhancement comes at the cost of a 1.06 W (166%) increase in pumping power, while improving the heat transfer coefficient by 1698.46 W/(m²·K) (33.35% enhancement).

Fig. 10 demonstrates that at $V_2 = 10.18$ L/min, the heat source surface exhibits improved temperature uniformity due to the significantly higher jet velocity. However, these significant advantages come at the cost of pressure drop, so it indicates great importance to explore more energy-efficient methods to improve the performance of heat sink.

3.5. Effect of jet hole arrangement

The above analysis reveals a hotspot near the center of Inlet 1. To eliminate the hotspots, jet orifices are added at the corresponding positions in the hot area, as shown by orifice plate 2 in Fig. 11(b). The

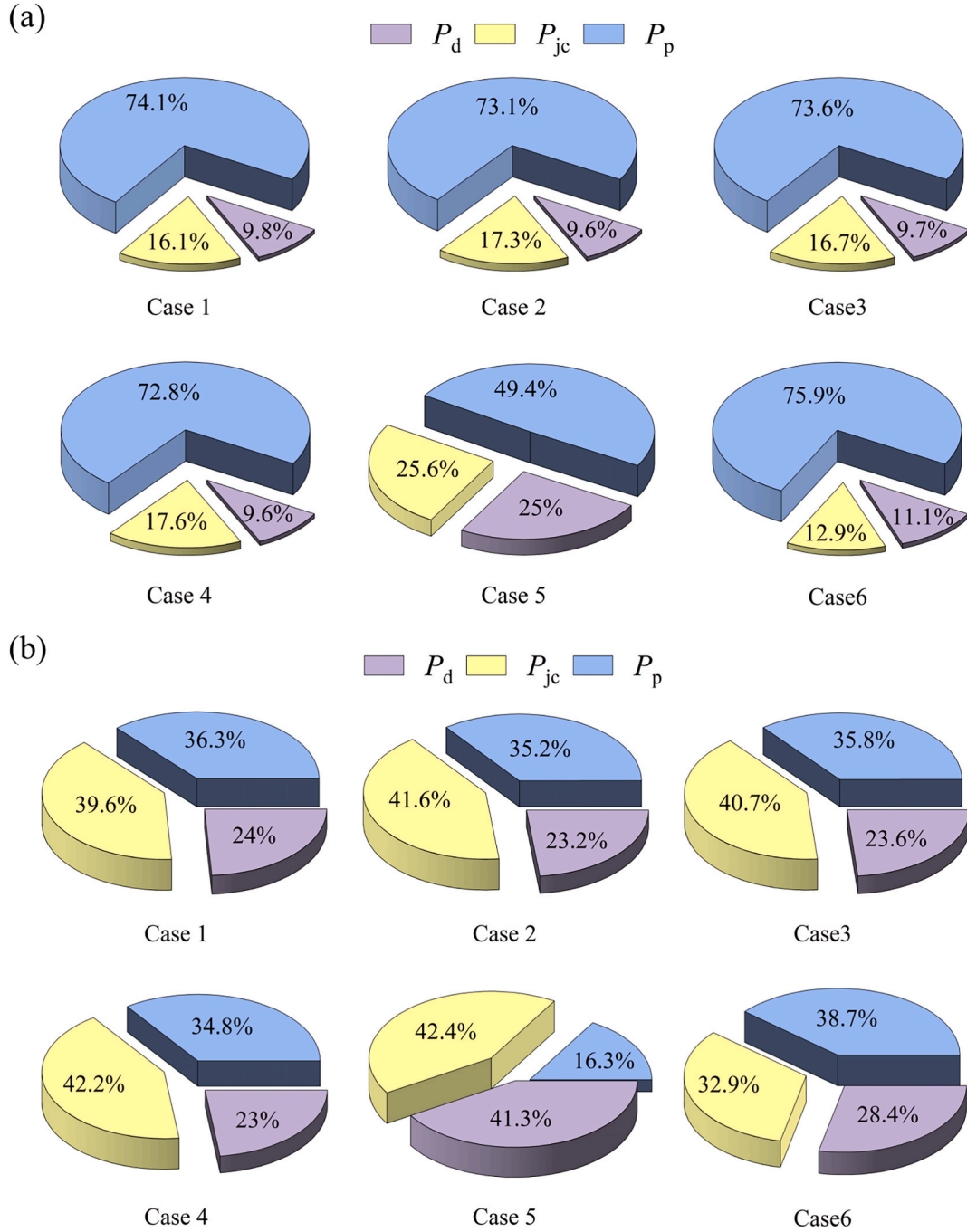


Fig. 14. The proportion of pumping power consumed by each component (a) $l = 1$ m, (a) $l = 0.2$ m.

original orifice arrangement (Plate 1) is also shown in Fig. 11(a) for comparison. The system configuration maintains a 2 mm jet impingement height with 25 W/cm² heat flux. A comparison is made between the effects of orifice Plate 1 and orifice Plate 2 on the heat sink's performance in this section.

The numerical simulation results are presented in Table 5. Overall, compared to the heat sink using Plate 1, the heat sink using Plate 2 exhibits a lower maximum temperature, a higher average temperature, a smaller temperature difference, a lower pumping power consumption, and a superior heat transfer coefficient. To be specific, using Plate 2 at 1.32 L/min modifies thermal performance as follows: average temperature rises 1.57 K while maximum temperature drops 4.52 K, temperature difference decreases 38.74%, heat transfer coefficient improves 9.67%, and pumping power reduces by 7.81%. At 2.2 L/min, Plate 2

demonstrates comparable performance enhancement, the heat source exhibits a 1.33 K increase in average temperature and a 3.47 K reduction in maximum temperature relative to Plate 1. The temperature difference decreases 36.5% (5.7 K reduction), while the heat transfer coefficient rises 7.6% and pumping power drops 9.1%. In summary, at lower flow rate, the heat sink with Plate 2 shows more pronounced thermal and pressure drop advantages.

Fig. 12(a) demonstrates that Plate 2 achieves more uniform temperature distribution compared to Plate 1, particularly at lower main-stream velocities. Notably, the localized hotspot near Inlet 1 observed with Plate 1 is eliminated when using Plate 2. Fig. 12(b) further explains the underlying mechanism of this observation, where the velocity enclosed by the black frame region of Plate 2 exceeds that of Plate 1 by a significant margin. The improved design of Plate 2 effectively suppresses

hotspot formation, resulting in more uniform temperature distribution across the entire heat source surface.

Fig. 13 presents the vorticity isosurface superimposed with the temperature field. The analysis reveals that the vortex structures have a significant impact on the temperature distribution of the heat source surface. Specifically, regions with vortices exhibit lower surface temperatures, while vortex-free regions show higher surface temperatures, particularly in the areas marked by the black wireframes. Moreover, large-scale vortices demonstrate a more pronounced cooling effect compared to small-scale vortices. Plate 1 generates larger but non-uniform vortex structures, leading to an uneven temperature distribution and higher thermal gradients. In contrast, Plate 2, equipped with three additional jet orifices, produces smaller yet more uniformly distributed vortices, resulting in a more uniform temperature distribution and reduced thermal gradients across the heat source surface. Similar trends are observed under different mainstream flow conditions.

3.6. Pumping power consumption analysis of immersed jet impingement for immersion cooling servers

This section analyzes the pumping power consumption ratio of different parts of the immersion jet impingement cooling system, aiming to provide practical guidance for engineering applications. A pipe is required in front of Inlet 2 of the distributor to introduce fluid. Using Equations 2.18 to 2.20, the pumping power in a 1 m long, 6 mm diameter pipe is calculated to be 4.885 W. When the length of the pipe is halved, the pumping power consumed in the pipe also decreases by half. In this section, six cases are selected, and their specific operating conditions are listed in Table 6. The proportion of pumping power consumed by each component for six cases are shown in Fig. 14 where P_d , P_{jc} and P_p denote the pumping power consumption by distributor, jet cavity and pipe, respectively. When the pipe length is 1 m, the pumping power in the pipe is dominant. Under the conditions, optimizing the fins, jet height, main flow rate, and jet orifices distribution little to the proportion of pumping power consumed by each component, whereas changing the jet flow rate significantly influences these proportions. When the pipe length is reduced to 0.2 m, the pumping power consumed by the pipe no longer dominates. In practical engineering applications, when the jet flow rate is high or long pipes are required, the optimization of the immersion jet cooling system has limited effect on reducing the pumping power. Therefore, in order to avoid a sharp increase in pumping power requirement, carefully regulate the jet flow rate and pipe length is essential.

4. Conclusions

Jet-assisted cooling enhances hotspot thermal management in immersion-cooled servers. A comprehensive investigation was conducted to evaluate the thermal performance and flow characteristics of the heat sink, examining the effects of fin structure modification, jet impingement height, mainstream/jet flow rates, and orifice distribution patterns. The following findings were summarized.

- (1) Intermittent fins outperform continuous fins regarding thermal performance and pressure drop due to the easier outflow of impingement liquid.
- (2) The optimal jet height are 2 mm and 2.5 mm and a smaller jet height can induce larger pressure drop, while a larger jet height leads to a smaller impingement velocity owing to the flow resistance of mainstream flow.
- (3) Within the studied range, increasing both the mainstream and jet flow rates enhances thermal performance, though at the expense of higher pumping power requirements.
- (4) The configuration of “4 × 4 + 3 × 1” jet orifices significantly improves temperature uniformity, lowers peak surface temperatures, and saves pumping power requirements.

- (5) In cases of high jet flow rate or extended pipe length, geometrical optimization on jet and heat sink has limited impact on reducing the total pumping power. It is crucial to regulate the jet flow rate and pipe length to prevent a sharp rise in pumping power.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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